Multi-Class Text Classification: A Comparison of Word Representations and ML/NN Models

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***Abstract*—In the era of unstructured data, efficient classification of question-answer pairs is vital for information retrieval systems. This project evaluates the performance of various machine learning and deep learning architectures on a multi-class text classification task. We compare traditional approaches, such as Multinomial Naive Bayes using TF-IDF, against sequence-based deep learning models such as Deep Neural Networks (DNN). We trained DNN using both Skip-gram and TF-IDF. Our project also includes models such as Simple RNNs, Gated Recurrent Units (GRU), and Long Short Term Memory (LSTM) networks and their their bidirectional variants. Feature extraction techniques involved TF-IDF and custom-trained Skip-gram Word2Vec embeddings. Furthermore, we conducted extensive hyperparameter tuning by comparing "Base" versions against "Tuned" versions equipped with dynamic embeddings, dropout regularization, and early stopping. Our results demonstrate that the Tuned Bi-GRU model achieved the highest accuracy of 70.38%, highlighting the importance of capturing sequential context and proper regularization in text classification.**

***Index Terms*—Text Classification, Deep Learning, Bi-GRU, Word2Vec, EDA, NLP.**

I. Introduction

Text classification is a fundamental task in Natural Language Processing (NLP), serving as the backbone for spam detection, sentiment analysis, and topic categorization. With the proliferation of community-based QA platforms (e.g., Yahoo Answers, Quora), the ability to automatically categorize content into domains such as "Sports," "Politics," or "Technology" is essential for user experience and content moderation.

This project focuses on the "Question Answer Classification Dataset 4," which presents a challenge common in real-world scraping: the data contains raw HTML tags and unstructured text. The objective is to develop a robust pipeline to clean, process, and classify this text into its respective categories.The dataset has 10 classes including Business Finance, Computers Internet, Education Reference, Entertainment Music, Family Relationships, Health, Politics Government, Science Mathematics, Society Culture and Sports.

We aim to benchmark the performance of traditional statistical models against modern recurrent neural networks. A key contribution of this study is a rigorous comparison between "Base" model architectures and "Hyperparameter Tuned" versions, specifically analyzing the impact of trainable word embeddings and regularization techniques on model convergence and generalization.

II. Methodology

A. Exploratory Data Analysis (EDA)

Before feeding data into any model, we conducted a thorough Exploratory Data Analysis to understand what we were working with. We visualized the distribution of the 10 target classes to check for imbalance, finding that the dataset was relatively well-balanced, which simplified our evaluation metric choice (accuracy). We also analyzed the length of the text samples. By plotting the word count distribution, we observed that most descriptions fell within a specific range, which helped us determine an appropriate maxlen (maximum sequence length) for padding our neural network inputs later.

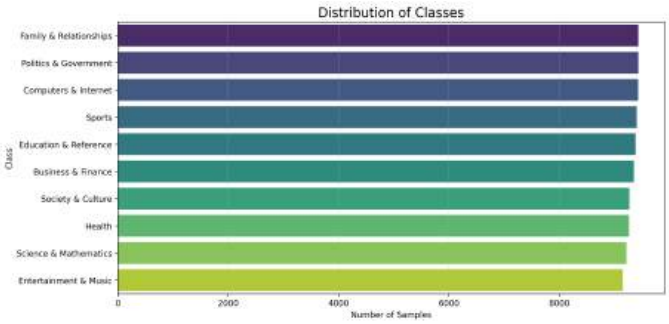


Fig. 1. Class Distribution of the Dataset.

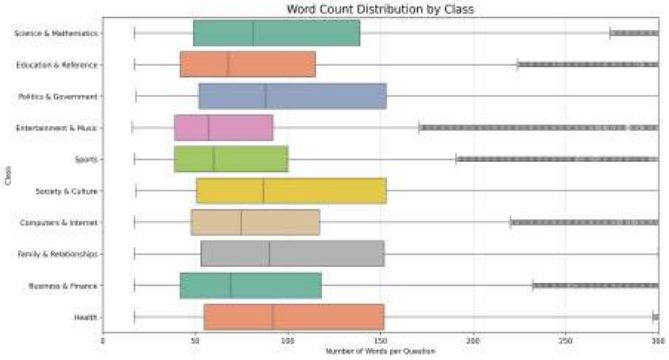


Fig. 2. Word Count.

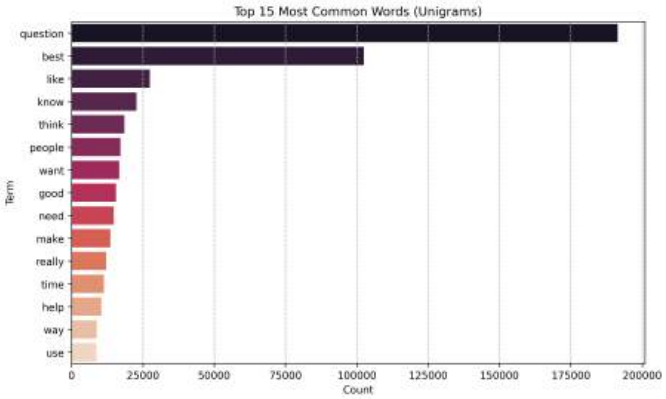


Fig. 3. Top Unigrams.

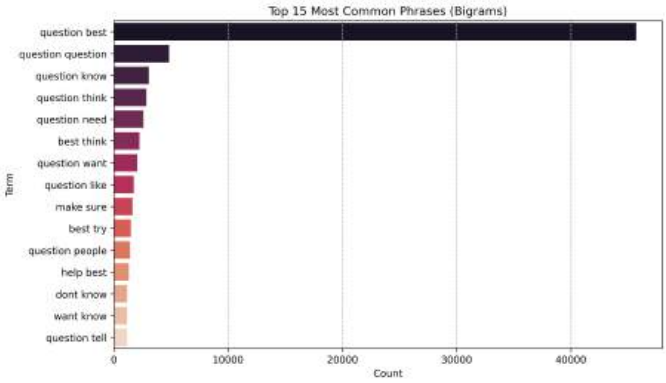


Fig. 4. Top Bigrams.

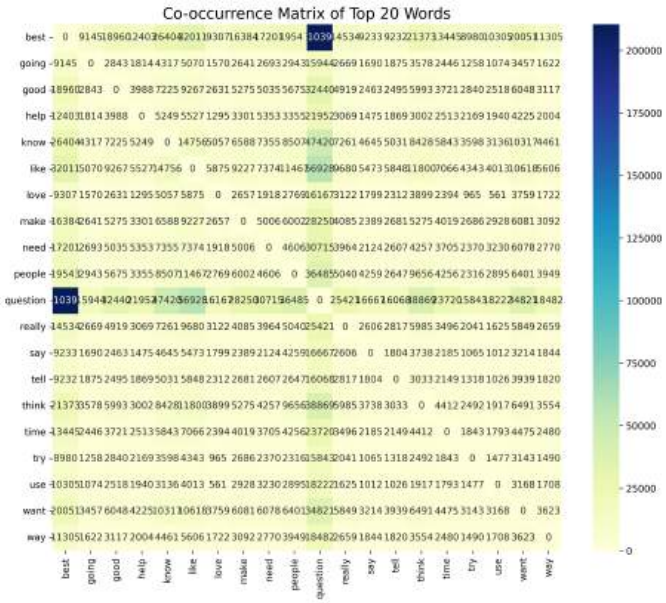




Fig. 6. Word Cloud.

B. Data Preprocessing and Word Representation

To convert raw text into clean, usable data, we built a comprehensive preprocessing pipeline using the NLTK and Python regular expression libraries. Our steps included:

- Noise Removal: We stripped out HTML tags, special characters, and numbers, keeping only alphabetic text.
- Normalization: All text was converted to lowercase to ensure words like ”Apple” and ”apple” were treated identically.
- Stopword Removal: We removed common English stopwords (like ”the”, ”is”, ”at”) that add noise without contributing significant meaning to the classification.
- Lemmatization: We applied WordNet Lemmatizer to reduce words to their base root form (e.g., ”running” becomes ”run”).
- Tokenization & Padding: For neural networks, we converted text into sequences of integers and padded them to a fixed length of 200 to allow for batch processing.

For RNN and Bi-RNN architectures, lemmatization and stopword removal were omitted to preserve interrogative signals. Feature extraction utilized custom Skip-gram Word2Vec embeddings.

C. Model Architectures

We implemented 9 distinct architectures. Each was evaluated in two configurations to measure the impact of hyperpa-

rameter optimization and regularization. For ML model, TF-IDF was used for training and for Neural Network Nodels, skip-gram was used. Additionally, DNN was trained using both.

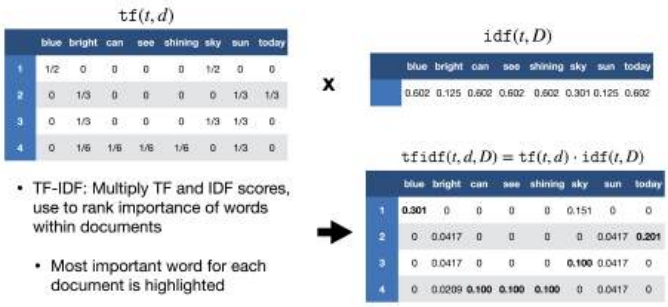


Fig. 7. Representation of TF-IDF.

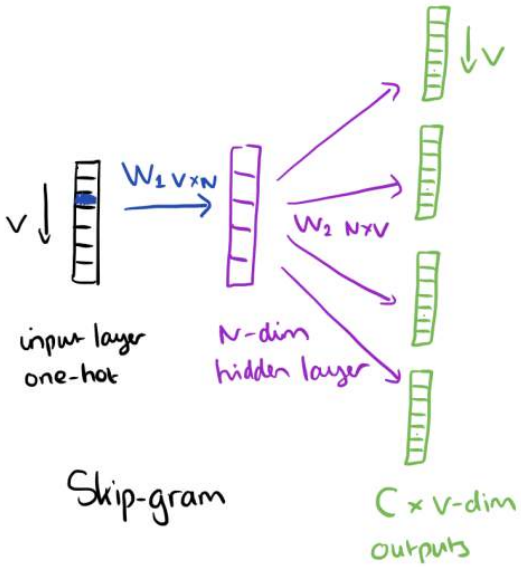


Fig. 8. Representation of Skip-gram.

1) Probabilistic and Dense Models:

- **Naive Bayes:** The *Base* version utilized default MultinomialNB. The *Tuned* version implemented *GridSearchCV* to find the optimal smoothing parameter α within $\{0.1, 1.0, 5.0, 10.0\}$.
- **DNN TF-IDF:** The *Base* model consisted of a single Dense layer (64 units). The *Tuned* model increased depth to a stacked architecture ($256 \rightarrow 128$), incorporating high Dropout rates (0.5, 0.3) to prevent over-reliance on sparse TF-IDF features.
- **DNN Skip-gram (SG):** *Base* used frozen 128-dim embeddings with Global Average Pooling (GAP) and a 64-unit Dense layer. *Tuned* utilized 200-dim **trainable** embeddings, a $128 \rightarrow 64$ dense stack, and Dropout.

2) Gated and Long-Term Recurrent Models: For GRU, Bi-GRU, LSTM, and Bi-LSTM, the transition from Base to Tuned followed a systematic upgrade:

- **Embedding Layer:** Base used static 128-dim vectors; Tuned used 200-dim trainable vectors.
- **Recurrent Depth:** Base models used a single 64-unit recurrent layer. Tuned models implemented stacked layers ($128 \rightarrow 64$) with *return_sequences=True* on the first layer and applied 0.2–0.3 dropout.
- **Optimization:** Tuned models utilized *EarlyStopping* (patience=5) and *ReduceLROnPlateau* (patience=3) to refine the learning process.

3) Simple and Bidirectional RNN Models: The RNN and Bi-RNN architectures were uniquely tuned to handle the sequential nature of raw text:

- **Simple RNN:** *Base* used 64 units with balanced class weights and a learning rate of $3e-4$. *Tuned* introduced *Recurrent Dropout* (0.2), an additional 32-unit Dense layer, and an increased learning rate of $5e-4$ over 30 epochs.
- **Bi-RNN:** *Base* used a Bidirectional wrapper over 64 units. *Tuned* increased unit capacity to 128, added recurrent dropout, and a post-recurrent 128-unit Dense layer with Dropout (0.2).

III. Results and Discussion

A. Quantitative Performance Metrics

Table I provides the accuracy and Macro F1-scores for all 18 configurations based on the experimental data.

TABLE I
Model Performance Comparison: Accuracy and Macro F1-Score

Model Name	Accuracy	F1 Macro
Naive Bayes (Base)	0.6669	0.6625
Naive Bayes (Tuned)	0.6686	0.6650
DNN TF-IDF (Base)	0.6355	0.6324
DNN TF-IDF (Tuned)	0.6747	0.6670
DNN SG (Base)	0.6695	0.6629
DNN SG (Tuned)	0.6809	0.6753
GRU (Base)	0.7002	0.6962
GRU (Tuned)	0.7002	0.6956
Bi-GRU (Base)	0.6990	0.6921
Bi-GRU (Tuned)	**0.7038**	**0.6970**
LSTM (Base)	0.6910	0.6840
LSTM (Tuned)	0.6850	0.6832
Bi-LSTM (Base)	0.6991	0.6939
Bi-LSTM (Tuned)	0.6999	0.6908
Simple RNN (Base)	0.5867	0.5898
Simple RNN (Tuned)	0.4730	0.4430
BIRNN (Base)	0.6358	0.6307
BIRNN (Tuned)	0.6326	0.6286

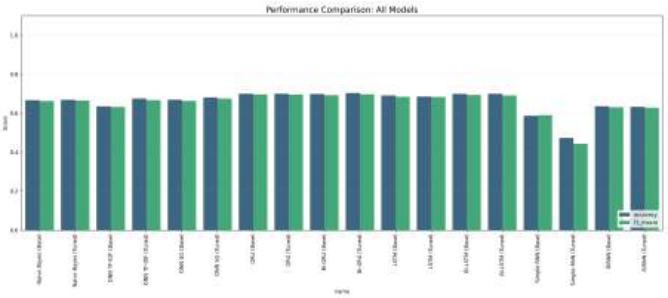


Fig. 9. Comparison of Performances

B. Identification of Best and Worst Performers

- **Best Performer (Tuned Bi-GRU):** Achieved 70.38% accuracy. The model's bidirectional nature allowed it to capture context from both directions, which is crucial for short text classification where the interrogative keyword can appear anywhere in the string.
- **Worst Performer (Tuned Simple RNN):** Surprisingly, the Tuned Simple RNN performed poorly (47.30%). This is attributed to the "vanishing gradient" problem inherent in simple RNNs. When additional depth and regularization were added, the model likely struggled to propagate gradients through the temporal layers, leading to significantly worse results than the baseline.

C. Comparison: ML Baseline vs. Best NN Model

A detailed comparison between the **Naive Bayes (ML)** and **Tuned Bi-GRU (NN)** reveals a clear architectural advantage:

- **Representation:** Naive Bayes relies on word counts/TF-IDF, treating words as independent features (Bag-of-Words). In contrast, the Bi-GRU uses dense word embeddings that capture semantic proximity (e.g., "CPU" and "Computer" are mathematically close).
- **Contextual Awareness:** Naive Bayes cannot distinguish between "Who is that person" and "That person is who," whereas the Bi-GRU preserves the sequence.
- **Performance Gap:** The Bi-GRU outperformed the ML baseline by nearly **4%** in accuracy. While Naive Bayes is robust for classes like *Sports*, it failed to distinguish nuanced overlaps in *Education* and *Science* that the Bi-GRU resolved through training.

D. Analysis of Base Model Confusion Matrices

An evaluation of the nine base confusion matrices reveals how different architectures perceive semantic boundaries in the dataset.

1) **Statistical and Dense Baselines:** Naive Bayes and DNN (TF-IDF) show strong diagonals for classes with distinct vocabularies, such as *Sports* and *Computers*. However, Naive Bayes exhibits significant "noise" in the *Society & Culture* category, often confusing it with *Family & Relationships* (569 instances).
2) **Recurrent Failure (RNN & BiRNN):** The Simple RNN and BiRNN base models show the weakest diagonals. The Simple RNN matrix shows a nearly uniform distribution of predictions, suggesting the model failed to converge due to vanishing gradients. BiRNN exhibits extremely low true-positive counts (e.g., only 67 correct for Business), highlighting the instability of basic recurrent units without gating.
3) **Gated Superiority (LSTM & GRU):** Gated models naturally outperformed simple RNNs. The GRU (Base) and LSTM (Base) correctly identified over 5,000 samples for *Computers & Internet*. However, a recurring trend across all gated models is the heavy confusion between *Education & Reference* and *Science & Mathematics* (averaging 900 misclassifications), likely due to shared academic terminology.

TABLE II
SUMMARY OF BASE MODEL ERROR PATTERNS

Category	Primary Strength	Primary Weakness
Statistical (NB)	High Accuracy in Sports	Class Overlap (Society/Family)
Simple Recurrent	Low latency	Failed to converge (Gradients)
Gated (LSTM/GRU)	Strong temporal tracking	Education vs. Science overlap
Bidirectional	Full context capture	High computational cost

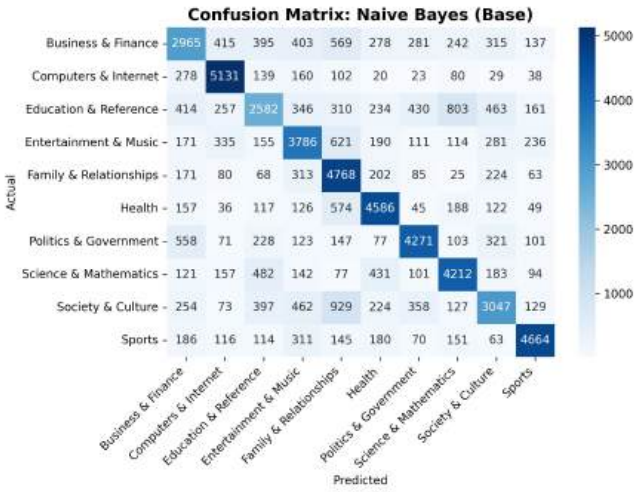


Fig. 10. Naive Bayes.

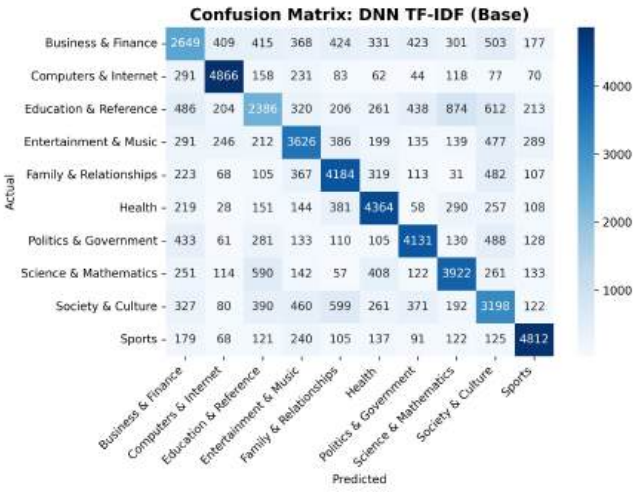


Fig. 11. Deep Neural Network(TD-IDF).

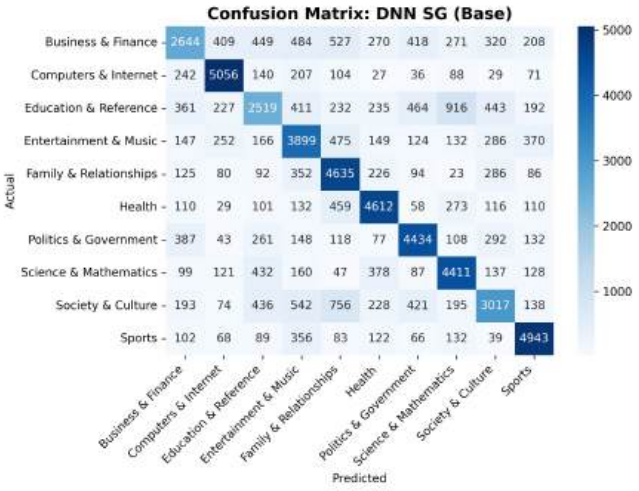


Fig. 12. Deep Neural Network(Skip-gram).

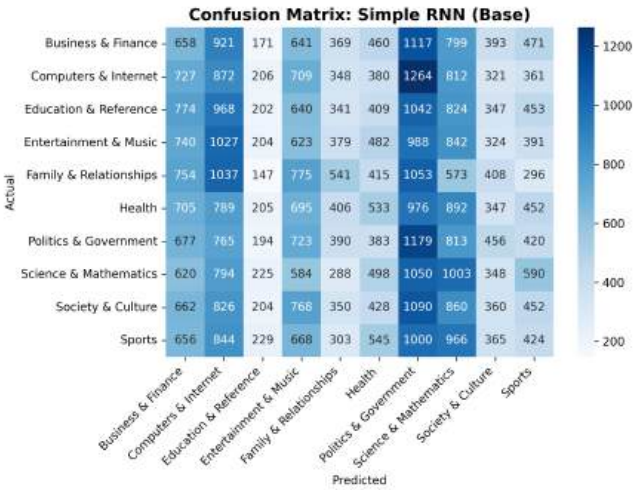


Fig. 13. Simple RNN.

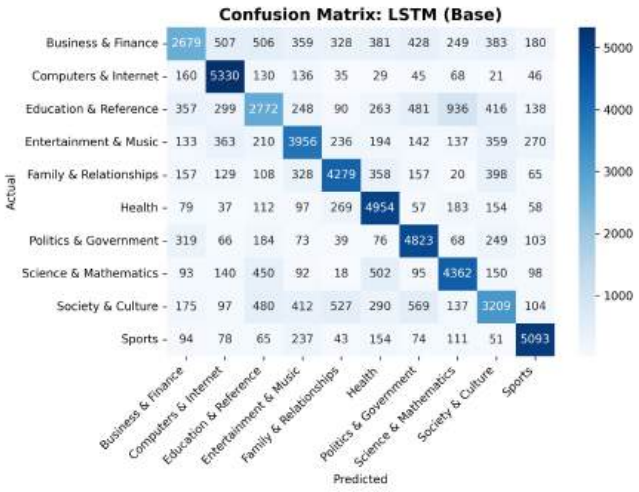


Fig. 14. LSTM.

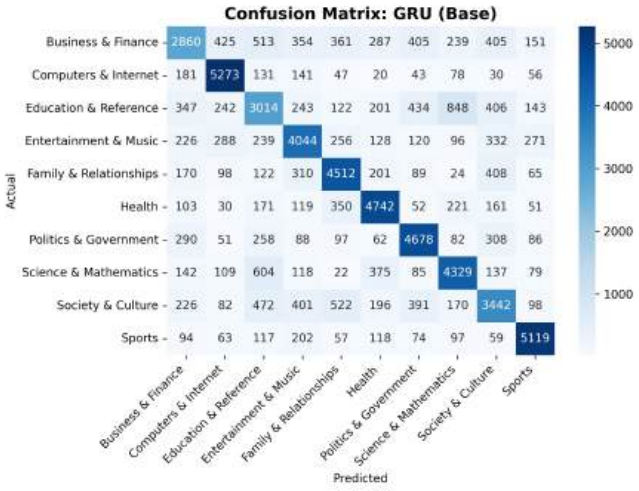


Fig. 15. GRU.

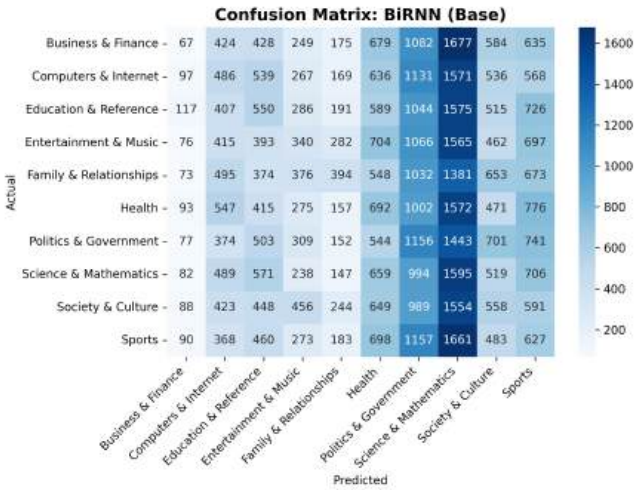


Fig. 16. BiRNN.

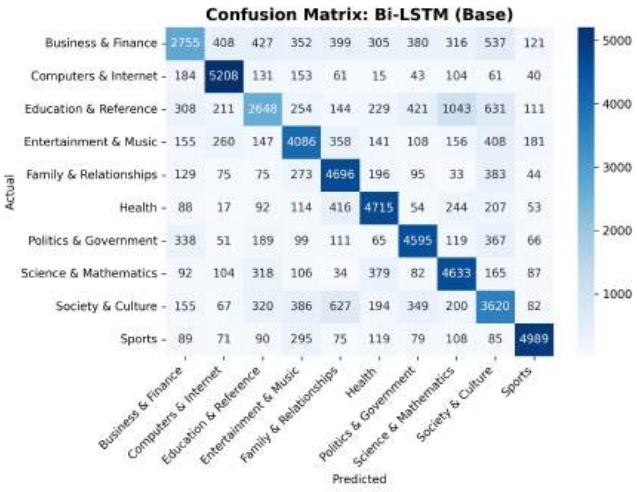


Fig. 17. BiLSTM.

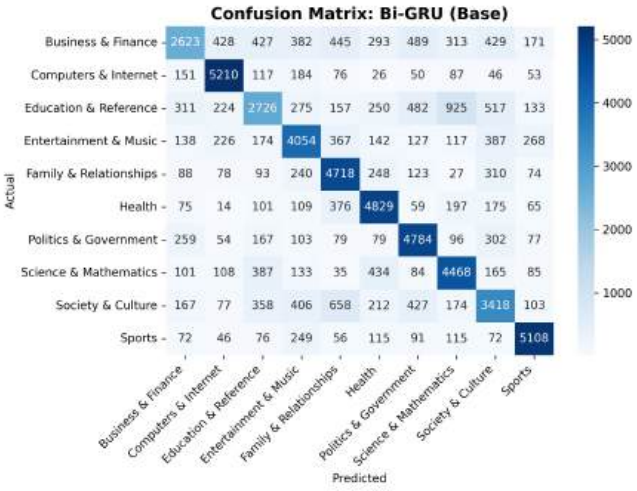


Fig. 18. BiGRU.

E. Superiority of the Best Performing Model

As shown in Table I, the **Tuned Bi-GRU** model achieved the highest accuracy (**70.38%**). This success is attributed to its bidirectional context tracking and the use of *trainable* 200-dim Skip-gram embeddings, which allowed the model to specifically differentiate between previously confused categories like Science and Education during the training process.

IV. Conclusion

A. Key Takeaways

Our experiments indicate that gated architectures like **Bi-GRU** and **Bi-LSTM** are significantly more robust for multi-class NLP tasks compared to simple recurrent units. The inclusion of trainable embeddings and dropout regularization were the two most influential factors in achieving a performance accuracy of **70.38%**.

B. Limitations

The primary limitations encountered were:

1) **Dataset Noise:** Despite cleaning, some HTML artifacts and semantic ambiguities between classes like "Education" and "Science" remained.
2) **Vanishing Gradients:** Non-gated models (RNN) were unable to learn long-term dependencies, leading to suboptimal performance in long-form questions.

C. Future Improvements

Future work should focus on utilizing **Transformer-based models** like BERT or RoBERTa. These models leverage self-attention mechanisms to understand the relationship between all words in a sentence simultaneously, which typically yields accuracies exceeding 85–90% on similar textual classification tasks.

References

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